



Zeek Cheat Sheet

A quick-reference guide for Zeek (formerly Bro): command-line usage, core logs, connection states, history letters, and key log fields for network detection and response.



Useful Zeek Commands

TASK	COMMAND
Analyze a PCAP	zeek -r capture.pcap
Live capture	zeek -i eth0
Ignore checksums	zeek -C -r file.pcap
Set the local network	zeek -r file.pcap "Site:local_nets ++ { 10.0.0.0/8 }"
Extract all files from PCAPs	zeek -C -r *.pcap /usr/local/share/zeek/policy/frameworks/files/extract-all-files.zeek
Extract transferred files	zeek -r file.pcap -e 'event file_new(f: fa_file) { Files::add_analyzer(f, Files::ANALYZER_EXTRACT); }'
Load policy scripts	zeek -r file.pcap policy/protocols/com/known-hosts
View the conn log	cat conn.log
Read compressed logs	zcat conn.log.gz less
Show connections as a table	zeek-cut ts uid id.orig_h id.orig_p id.resp_h id.resp_p proto conn_state < conn.log
List unique originators	zeek-cut id.orig_h < conn.log sort -u
Count connections	wc -l conn.log
Filter DNS queries	zeek-cut ts query qtype name rcode_name < dns.log head
HTTP status failures	zeek-cut ts method host url status_code < http.log awk '\$5 >= 400'
Find notices	grep Notice notice.log head
Deploy via zeekctl	zeekctl deploy

Command-Line Usage

COMMAND	PURPOSE
grep -H -P '*#fields' *.log	List fields in all log files.
zeek -r capture.pcap	Analyze a pcap file offline (writes logs).
zeek -i eth0	Capture and analyze live traffic on an interface.
zeek -C -r file.pcap	Ignore invalid IP/TCP checksums.
zeek -e 'event ...'	Run a Zeek event expression directly.
zeek -s signatures.sig	Load a signatures file.
zeek -l dir	Write logs to a local directory.
zeek -N	List loaded modules / analyzers.
zeek-cut field1 field2 ... < log	Extract specific fields from a log file.
zeek-cut -d	Convert time values into human readable format
zeek-cut -n	Print all fields "except" those specified (negate selection).
zeek-cut -u	Similar to -d, but print timestamps in UTC instead of local.

Core Logs Reference

LOG	WHAT IT LOGS
conn.log	IP, TCP, UDP, and ICMP connections.
dns.log	DNS queries and responses.
http.log	HTTP requests and replies.
ssl.log	TLS/SSL handshakes, ciphers, and cert.
x509.log	X.509 certificate details and fingerprints.
files.log	File analysis results (MIME, hashes).
notice.log	Interesting events and activity (alerts).
weird.log	Unexpected or anomalous activity (alerts).
ssh.log	SSH handshakes and authentication
dhcp.log	DHCP lease activity.
ftp.log	FTP commands and replies.
smtp.log	SMTP email transactions.
ntlm.log	NTLM authentication.
kerberos.log	Kerberos authentication (AS/TGS).
rdp.log	Remote Desktop Protocol activity.
pe.log	Portable executable metadata.
syslog.log	Syslog messages.
dce_rpc.log	DCE/RPC calls.
radius.log	RADIUS authentication attempts.
sip.log	SIP calls.
snmp.log	SNMP messages.
socks.log	SOCKS proxy requests.
mysql.log	MySQL commands.
irc.log	IRC communication.
ldap.log	LDAP transactions.
quic.log	QUIC connection updates.
tunnel.log	Encapsulating tunnels (GRE, IP-in-IP, etc.).
websocket.log	WebSocket handshakes.
analyzer.log	Protocol analysis violations.

http.log

FIELD	DESCRIPTION
ts	Time of the request
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
trans_depth	Pipelined request depth
method	HTTP verb
host	Host header
uri	Request URI
referrer	Referer header
version	Reply HTTP version
user_agent	User-Agent header
origin	Origin header
request_body_len	Uncompressed client body size
response_body_len	Uncompressed server body size
status_code	Server status code
status_msg	Server status message
info_code	Last 1xx informational code
info_msg	Last 1xx informational message
tags	Discovered attributes
username	Client username
password	Captured password
proxied	Proxy-related headers
orig_fuids	Client file UIDs
orig_filenames	Client filenames
orig_mime_types	Client MIME types
resp_fuids	Server file UIDs
resp_filenames	Server filenames
resp_mime_types	Server MIME types

sip.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
trans_depth	Pipelined transaction depth
method	SIP verb
uri	Request URI
date	Date header
request_from	Request From header
request_to	Request To header
response_from	Response From header
response_to	Response To header
reply_to	Reply To header
call_id	Call-ID
seq	CSeq
subject	Subject header
request_path	Request Via path
response_path	Response Via path
user_agent	User-Agent
status_code	Status code
status_msg	Status message
warning	Warning header
request_body_len	Request Content-Length
response_body_len	Response Content-Length
content_type	Server Content-Type

dns.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
proto	Transport protocol
trans_id	DNS transaction ID
rtt	Query-to-response RTT
qtype	Queried domain name
qclass	Query class value
qclass_name	Query class name
qtype	Query type value
qtype_name	Query type name
rcode	Response code value
rcode_name	Response code name
AA	Authoritative answer
TC	Truncated
RD	Recursion desired
RA	Recursion available
Z	Reserved Z field
answers	Answer resource records
TTLs	Caching intervals of answers
rejected	Query rejected by server
opcode	DNS opcode value
opcode_name	DNS opcode name

files.log

FIELD	DESCRIPTION
ts	Time file was first seen
fuid	File UID
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
source	Analyzer that produced the file
depth	Encapsulation depth
analyzers	File analyzers applied
mime_type	Detected MIME type
filename	Filename from the source
local_orig	Originator is local
is_orig	File sent by originator
seen_bytes	Bytes seen by analysis
total_bytes	Expected full file size
missing_bytes	Bytes missed (packet loss)
overflow_bytes	Bytes not delivered to stream analyzers
timeout	Analysis timed out
parent_fuid	Parent container file UID
md5	MD5 digest
sha1	SHA1 digest
sha256	SHA256 digest
extracted	Extracted local filename
extracted_cutoff	Extraction was truncated
extracted_size	Bytes written to disk

smtp.log

FIELD	DESCRIPTION
ts	Time message first seen
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
trans_depth	Message transaction depth
helo	HELO/EHLO contents
mailfrom	MAIL FROM address
rcptto	RCPT TO addresses
date	Date header
from	From header
to	To header
cc	CC header
reply_to	Reply-To header
msg_id	Message-ID
in_reply_to	In-Reply-To
subject	Subject
x_originating_ip	X-Originating-IP
first_received	First Received header
second_received	Second Received header
last_reply	Last server reply
path	Transmission path from headers
user_agent	User-Agent
tls	Connection switched to TLS
fuids	Attached file UIDs

conn.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
proto	Transport protocol
service	Detected application protocol(s)
duration	Connection duration
orig_bytes	Payload bytes from originator
resp_bytes	Payload bytes from responder
conn_state	Connection state
local_orig	Originator is local
local_resp	Responder is local
missed_bytes	Bytes lost to content gaps
history	Connection history flags
orig_pkts	Packets from originator
orig_ip_bytes	IP-layer bytes from originator
resp_pkts	Packets from responder
resp_ip_bytes	IP-layer bytes from responder
tunnel_parents	Parent tunnel connection UIDs
ip_proto	IP header protocol number

rdp.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
cookie	Client cookie (often truncated username)
result	Negotiation/GCC status
security_protocol	Protocol chosen by server
client_channels	Channels requested by client
keyboard_layout	Client keyboard language
client_build	RDP client version
client_name	Client machine name
client_dig_product_id	Client product ID
desktop_width	Client desktop width
desktop_height	Client desktop height
requested_color_depth	Requested color depth
cert_type	Native RDP cert type
cert_count	Native RDP cert count
cert_permanent	Cert is permanent
encryption_level	Encryption level
encryption_method	Encryption method

ssl.log

FIELD	DESCRIPTION
ts	Time SSL first detected
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
version	SSL/TLS version server chose
cipher	Cipher suite server chose
curve	ECDH/ECDHE curve if used
server_name	SNI hostname from client
resumed	Session resumed prior keys
last_alert	Last TLS alert seen
next_protocol	ALPN protocol server chose
established	Handshake completed successfully
ssl_history	Ordered handshake message codes
cert_chain_fps	Server cert fingerprint chain
client_cert_chain_fps	Client cert fingerprint chain
subject	Server certificate subject
issuer	Server certificate issuer
client_subject	Client certificate subject
client_issuer	Client certificate issuer
sni_matches_cert	SNI matches server cert

kerberos.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
request_type	AS, TGS, or AP
client	Client principal
service	Service principal
success	Request succeeded
error_msg	Error message
from	Ticket valid from
till	Ticket valid until
cipher	Ticket encryption type
forwardable	Forwardable flag requested
renewable	Renewable flag requested
client_cert_subject	Client cert subject
client_cert_fuid	Client cert file UID
server_cert_subject	Server cert subject
server_cert_fuid	Server cert file UID

ftp.log

FIELD	DESCRIPTION
ts	Time when the command was sent
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
user	FTP session username
password	FTP password if captured
command	Client FTP command
arg	Command argument
mime_type	Sniffed file MIME type
file_size	Transfer size if known
reply_code	Server reply code
reply_msg	Server reply message
data_channel_passive	Data channel uses PASV/EPSV
data_channel_orig_h	Data channel initiator IP
data_channel_resp_h	Data channel acceptor IP
data_channel_resp_p	Data channel acceptor port
fuid	File unique ID

ntp.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
version	NTP version
mode	NTP mode
stratum	Stratum
poll	Max interval between messages
precision	System clock precision
root_delay	Delay to reference clock
root_disp	Dispersion to reference clock
ref_id	Reference clock ID
ref_time	Last clock set time
org_time	Client depart time
rec_time	Server receive time
xmt_time	Server transmit time
num_exts	Unparsed extension field count

smb_files.log

FIELD	DESCRIPTION
ts	Time file first discovered
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
fuid	File UID
action	SMB file action
path	Tree path
name	Filename
size	Total file size
prev_name	Previous name (rename)
times.modified	Modification time
times.accessed	Access time
times.created	Creation time
times.changed	Change time

dhcp.log

FIELD	DESCRIPTION
ts	Time of earliest DHCP message
uids	Connection UIDs in the conversation
client_addr	Client IP
server_addr	Server IP
mac	Client hardware address
host_name	Hostname option (12)
client_fqdn	Client FQDN option (81)
domain	Domain name option (15)
requested_addr	Requested IP
assigned_addr	Assigned IP
lease_time	Lease interval
client_message	Client DECLINE/NAK reason
server_message	Server NAK reason
msg_types	DHCP message types in the transaction
duration	First-to-last message time

irc.log

FIELD	DESCRIPTION
ts	Time command was seen
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
nick	IRC nickname
user	IRC username
command	Client IRC command
value	Command value/target
addl	Additional command data
dcc_file_name	DCC filename requested
dcc_file_size	DCC size claimed by sender
dcc_mime_type	Sniffed DCC MIME type
fuid	File unique ID

ldap_search.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
message_id	LDAP message ID
scope	Search scope
deref_aliases	Alias dereference policy
base_object	Search base DN
result_count	Number of results
result	Result code
diagnostic_message	Server diagnostic
filter	Search filter
attributes	Requested attributes

postgresql.log

FIELD	DESCRIPTION
ts	Time of activity
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
user	User from StartupMessage
database	Database from StartupMessage
application_name	App name from StartupMessage
frontend	Frontend/client message type
frontend_arg	Frontend message argument
backend	Backend/server message type
backend_arg	Backend message argument
success	Whether the request succeeded
rows	Rows returned if applicable

snmp.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
duration	First-to-latest packet
version	SNMP version
community	v1/v2c community or v3 username
get_requests	GET varbind count
get_bulk_requests	GETBULK varbind count
get_responses	GET response varbind count
set_requests	SET varbind count
display_string	Responder sysDescr
up_since	Claimed boot time

radius.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
username	Username
mac	MAC address
framed_addr	Assigned address hint
tunnel_client	Tunnel-Client-Endpoint
connect_info	Connect-Info
reply_msg	Server challenge/reply text
result	Success or failure
ttl	Time from first request to Accept/error

ldap.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
message_id	LDAP message ID
version	LDAP version
opcode	LDAP operation
result	Result code
diagnostic_message	Server diagnostic
object	Target object
argument	Operation argument

ntlm.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
username	Username from client
hostname	Hostname from client
domainname	Domain from client
server_nb_computer_name	Server NetBIOS name (challenge)
server_dns_computer_name	Server DNS name (challenge)
server_tree_name	Server tree name (challenge)
success	Authentication succeeded

quic.log

FIELD	DESCRIPTION
ts	Time of first QUIC packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
version	QUIC version from client INITIAL
client_initial_dcid	Client first destination CID
server_scid	Server-chosen source CID
server_name	SNI from ClientHello if seen
client_protocol	First ALPN from ClientHello
history	QUIC packet-type history string

dce_rpc.log

FIELD	DESCRIPTION
ts	Time of first packet
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
rtt	Request-to-response RTT
remote_pipe	Remote pipe name
endpoint	Endpoint name (from UUID)
operation	RPC operation

smb_mapping.log

FIELD	DESCRIPTION
ts	Time tree was mapped
uid	Connection UID
id.orig_h	Originator IP
id.orig_p	Originator port
id.resp_h	Responder IP
id.resp_p	Responder port
path	Tree path
service	Resource type (disk/printer/pipe)
native_file_system	Tree filesystem
share_type	SMB2 share type

tunnel.log

FIELD	DESCRIPTION
ts	Time of tunnel activity
uid	Tunnel UID (if applicable)
id.orig_h	Tunnel originator IP
id.orig_p	Tunnel originator port (0 if IP-in-IP)
id.resp_h	Tunnel responder IP
id.resp_p	Tunnel responder port (0 if IP-in-IP)
tunnel_type	Encapsulation type
action	DISCOVER, CLOSE, or EXPIRE

weird.log